deflection = ... mm [3]

[Total: 8]

7 A 240 V a.c. mains is connected to a step-down transformer, a full wave rectifier and a motor as shown in Fig. 7.1. The electric motor requires an operating peak power of 6.6 W.

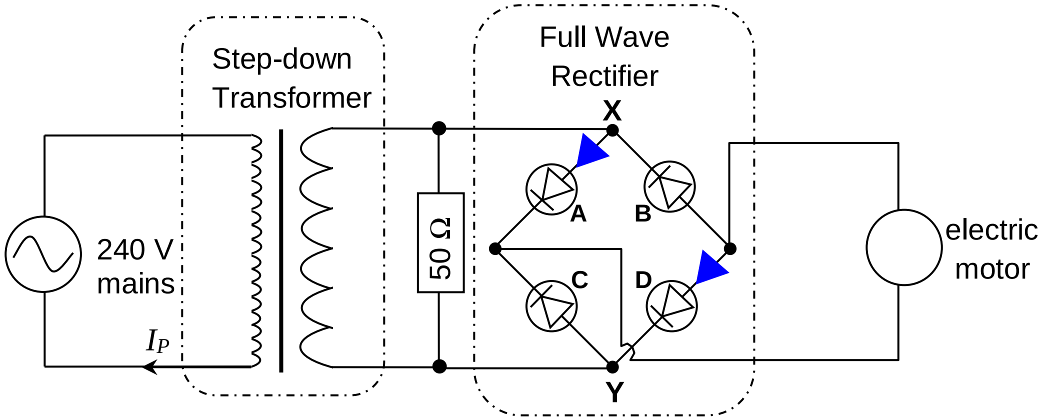

Fig. 7.1

The variation of potential difference across the 50 Ω resistor with time is shown in Fig. 7.2.

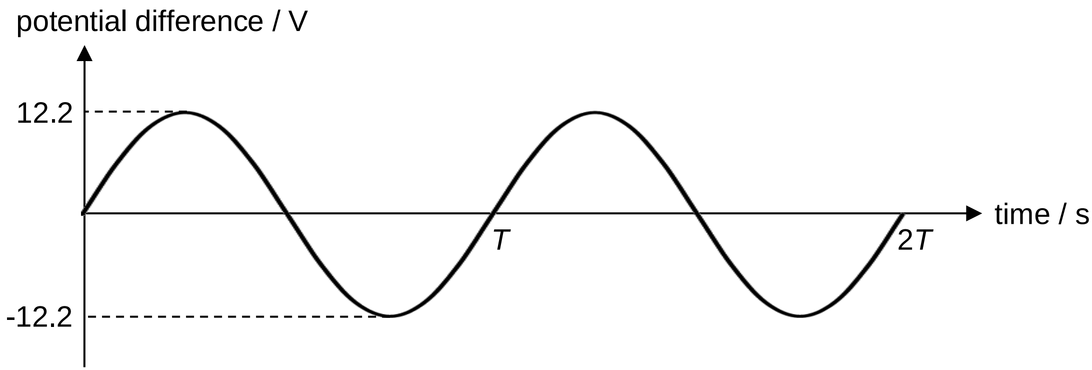

Fig. 7.2

(a) Determine the turns ratio of primary coil to secondary coil in the step-down transformer.

For sinusoidal input, peak $V_P = 240\sqrt{2} = 339\text{ V}$

$$\therefore \frac{N_P}{N_S} = \frac{V_P}{V_S} = \frac{339}{12.2} = 27.8$$

turns ratio = ... [2]

(b) (i) For half a period T, the potential at X is higher than that at Y.

On Fig. 7.1, draw an arrow to indicate the direction of current, if any, across the four diodes A, B, C and D during this half of a period. [1]

(ii) Sketch, on Fig. 7.3, the variation of potential difference across the electric motor with time. Numerical calculation is not required. [1]

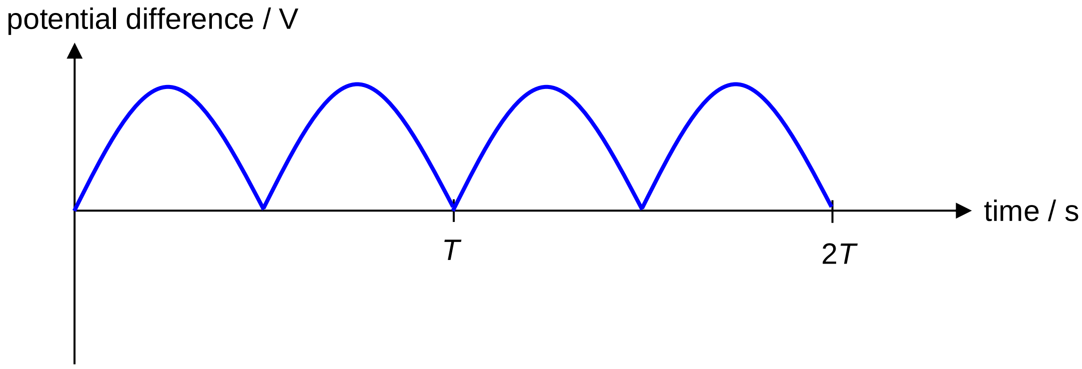

Fig. 7.3

(iii) Hence, sketch, on Fig. 7.4, the variation of output power of the step-down transformer with time when the electric motor is operating at its normal operating power. Include appropriate values in the output power axis.

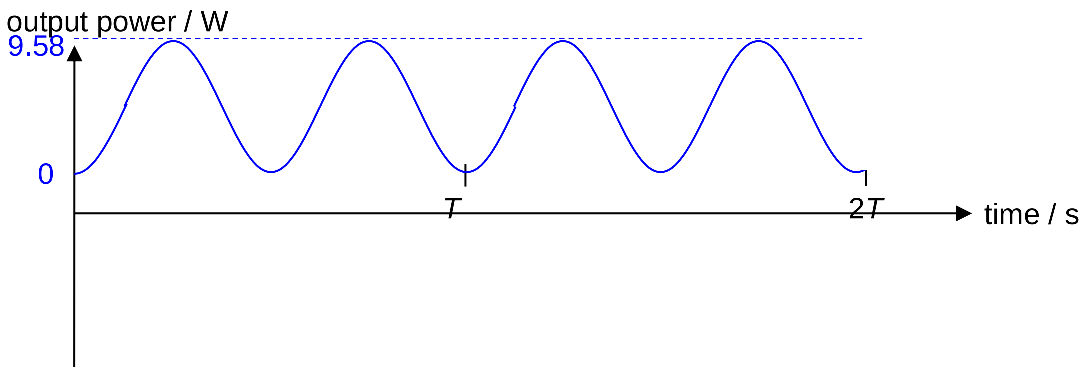

Fig. 7.4

Total peak power output required from transformer

= P_o across motor + P_o across 50 Ω resistor

$$= 6.6 + \frac{(12.2)^2}{50} = 9.58\text{ W}$$

[2]

(c) Given that the transformer is 80% efficient, determine the r.m.s. current flowing through the primary coil.

$<P> = \frac{1}{2} P_o = 4.7884$ W

Since transformer is 80% efficient, $0.8 \times < P_{in} > = < P_{out} >$

$$0.8 \times 240 \times I_{Pri,rms} = 4.7884$$

$$I_{Pri} = 0.0249 \text{ A}$$

current = ... A [2]

[Total: 8]

8 In a refrigeration system, a fluid known as a refrigerant undergoes alternating changes in its